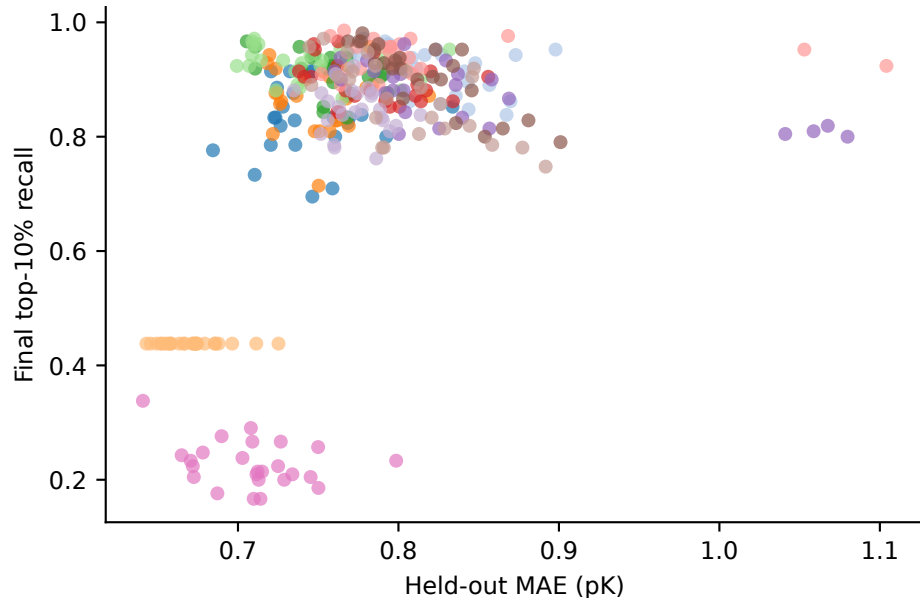
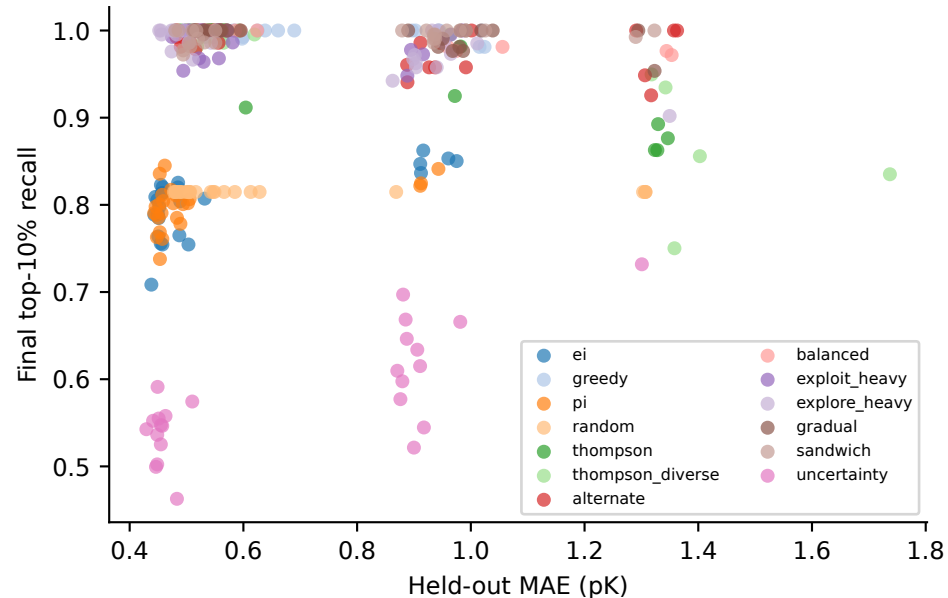
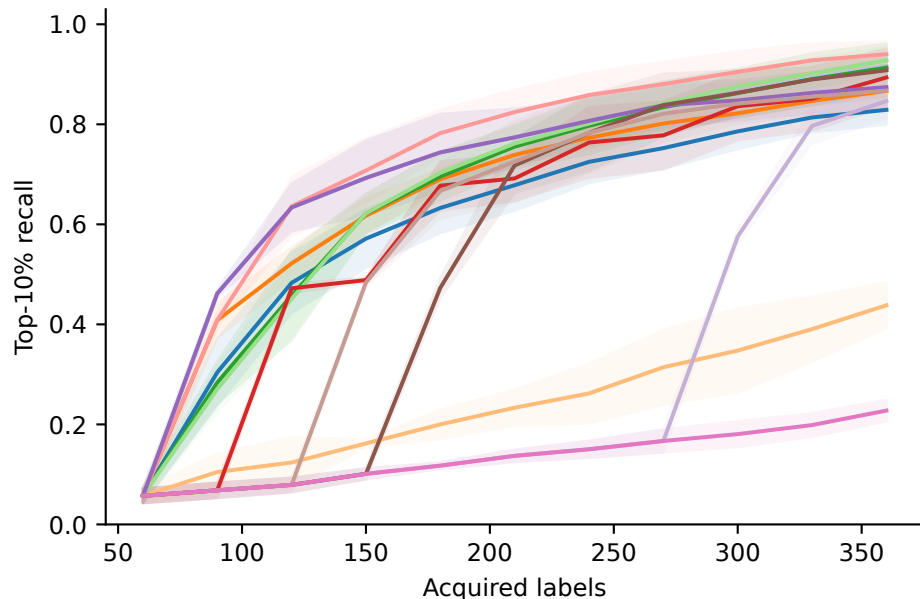
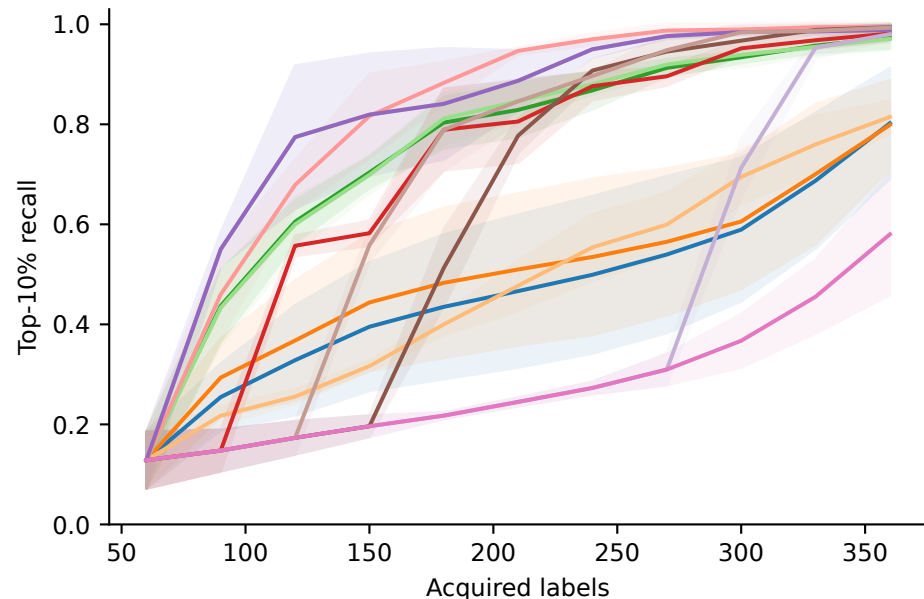
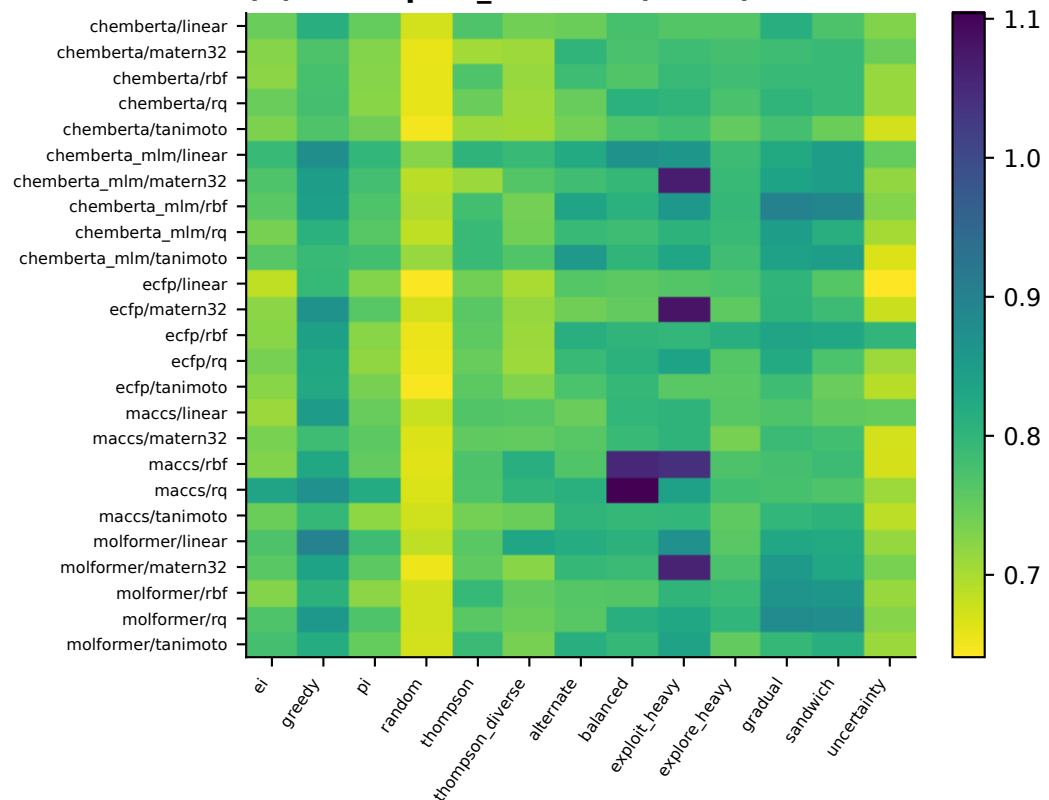
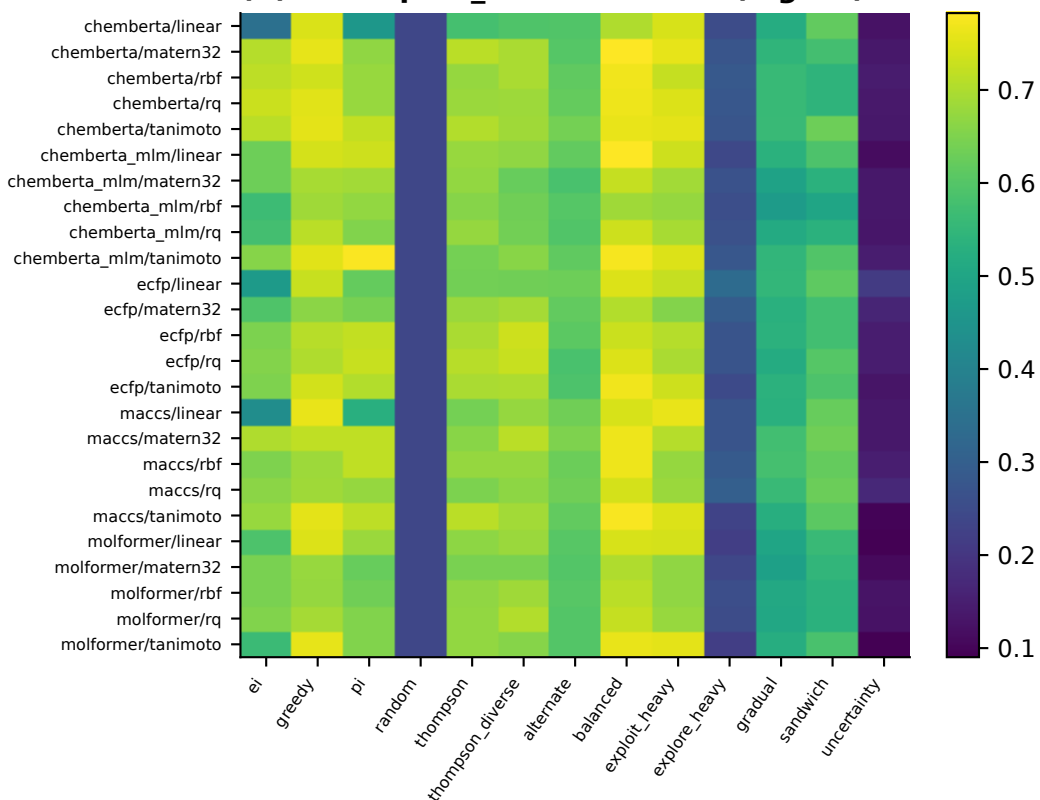
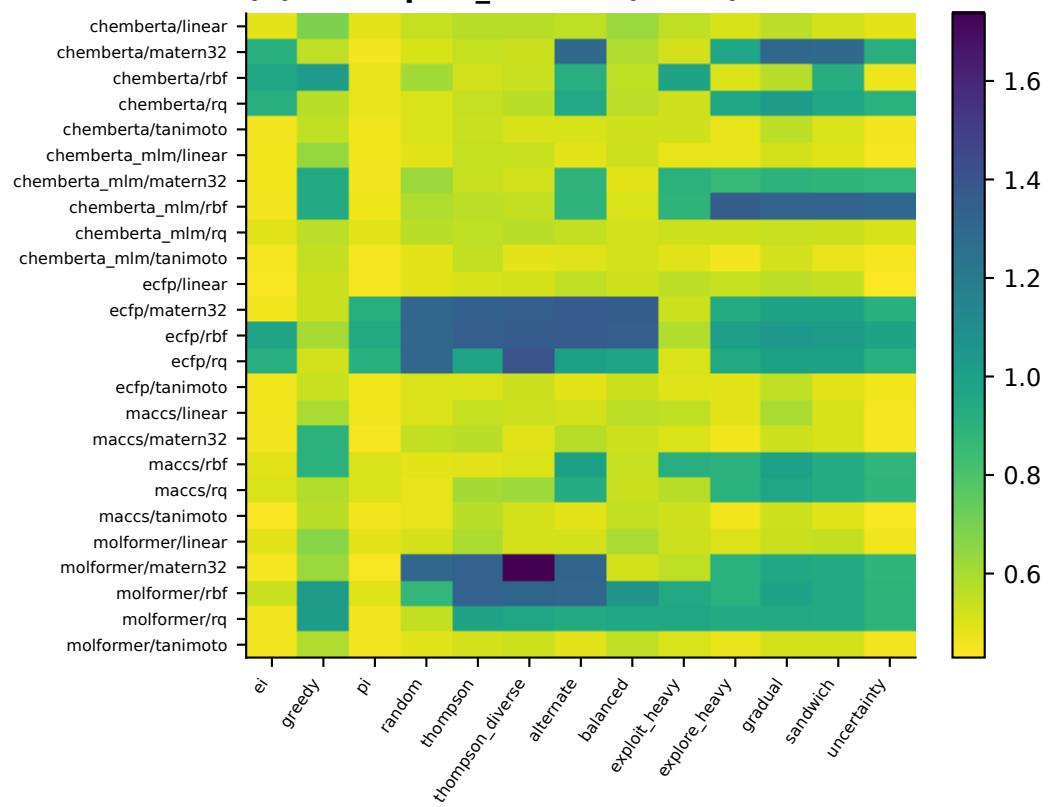
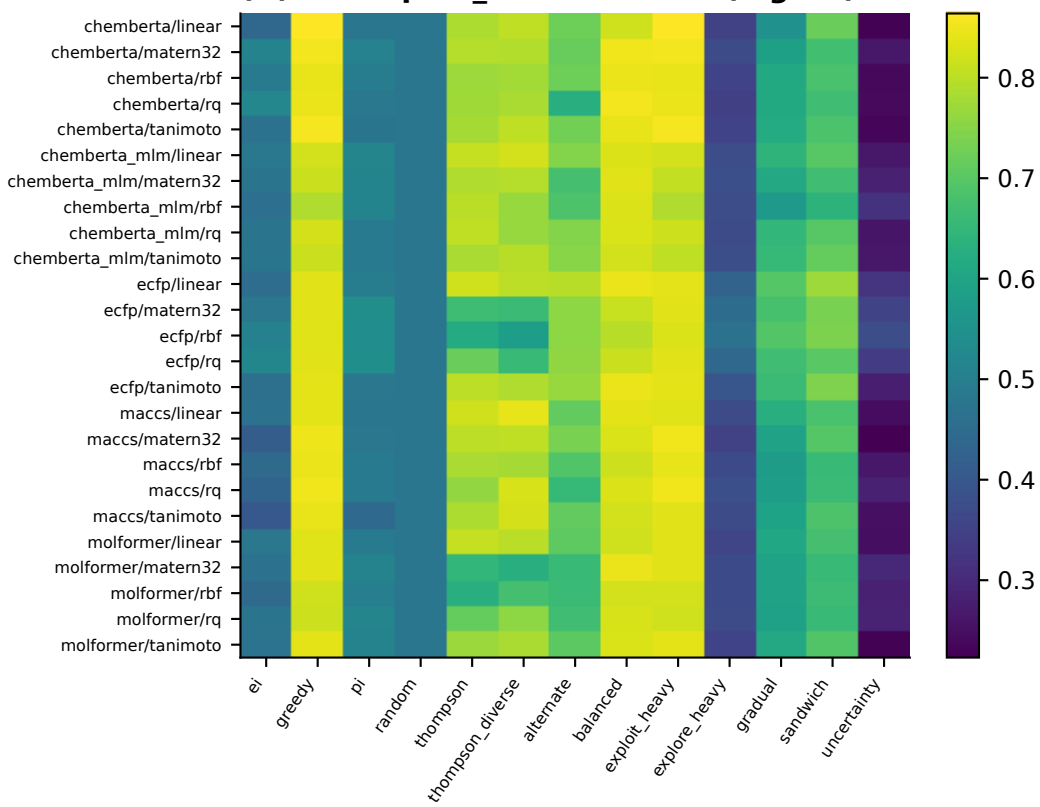
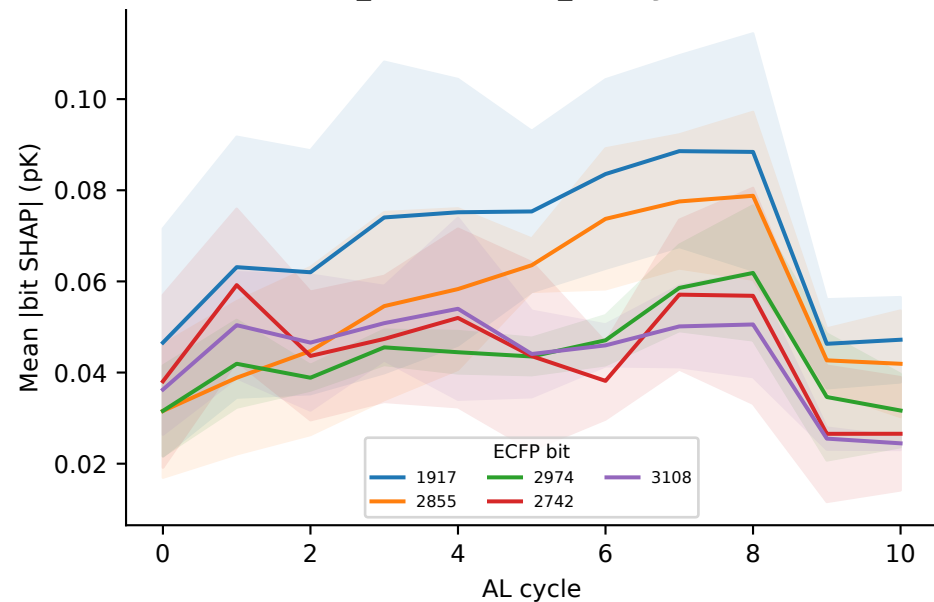
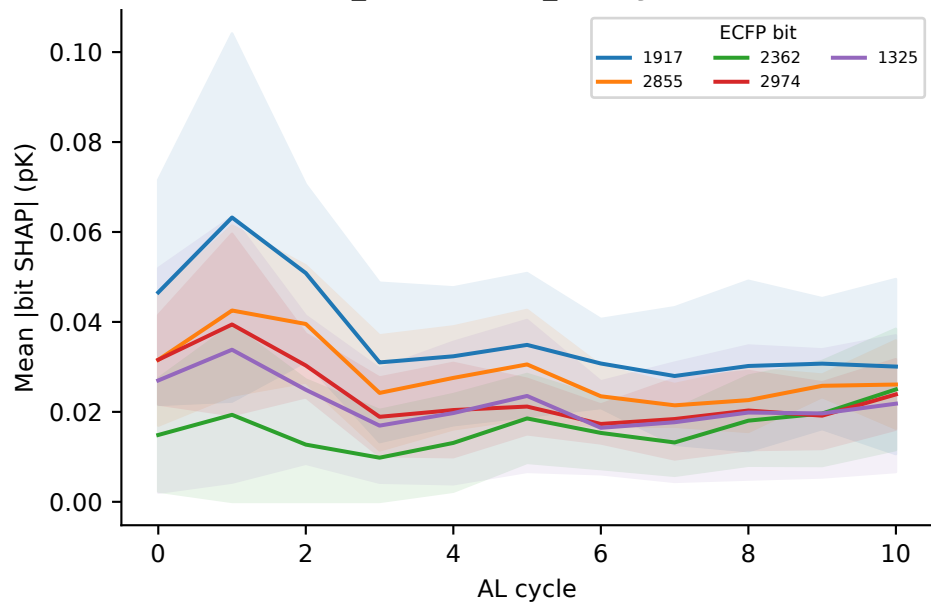
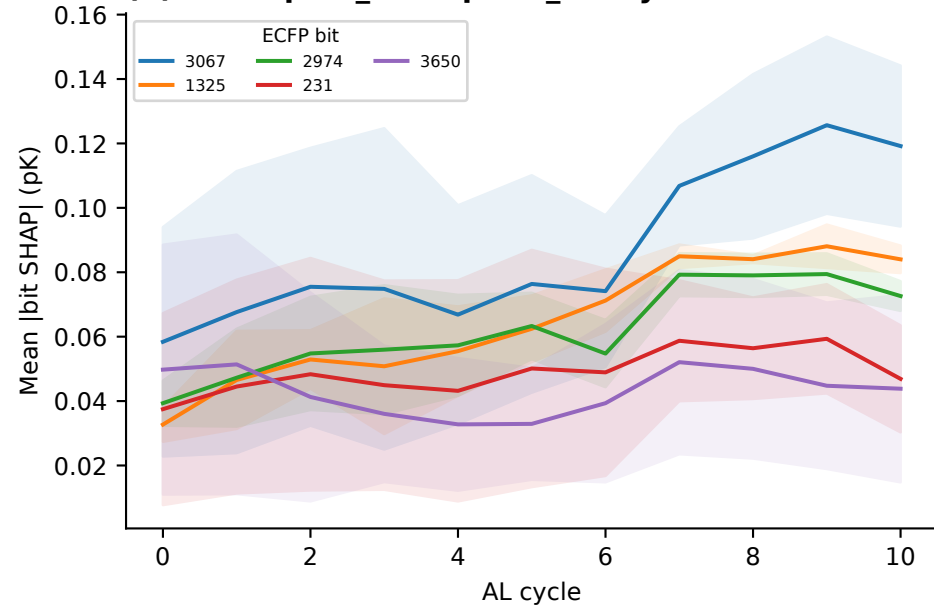
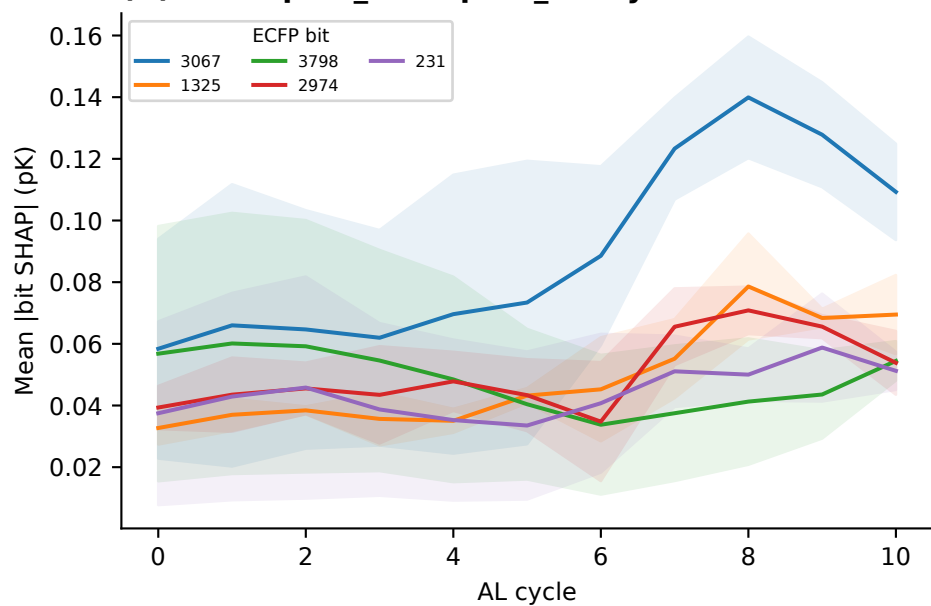
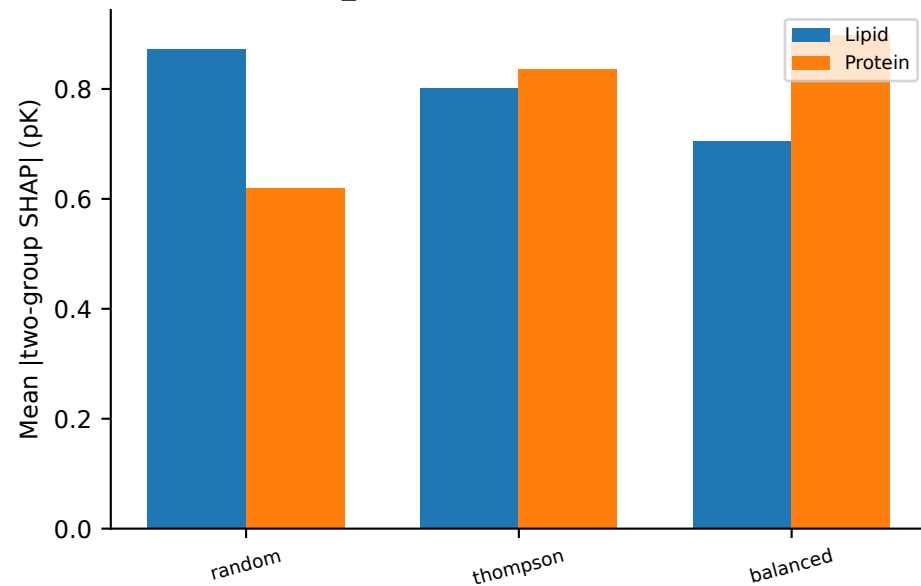
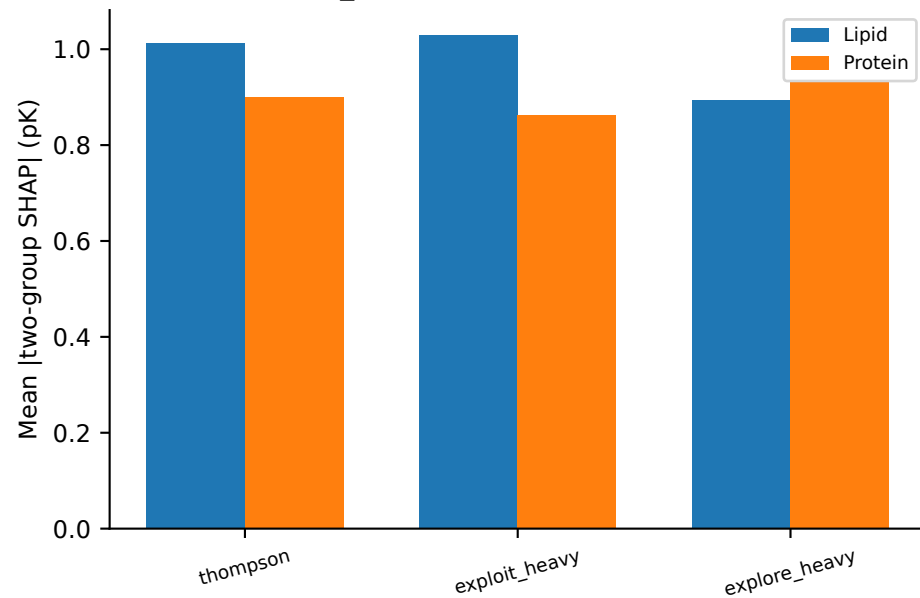
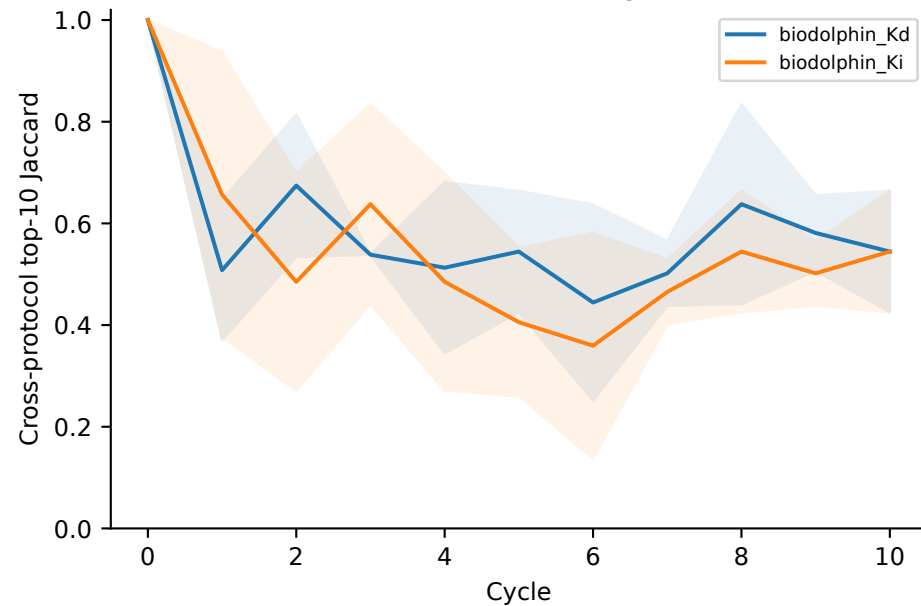
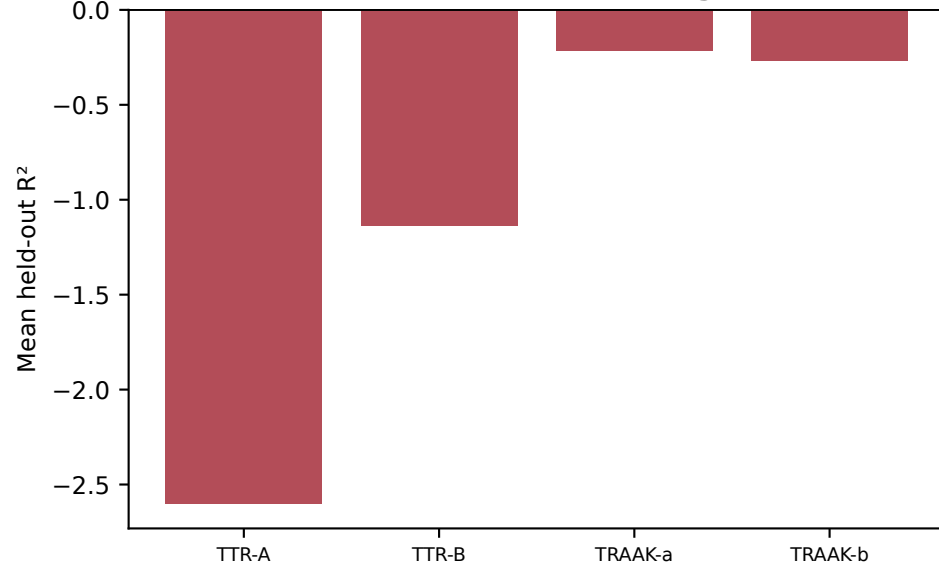
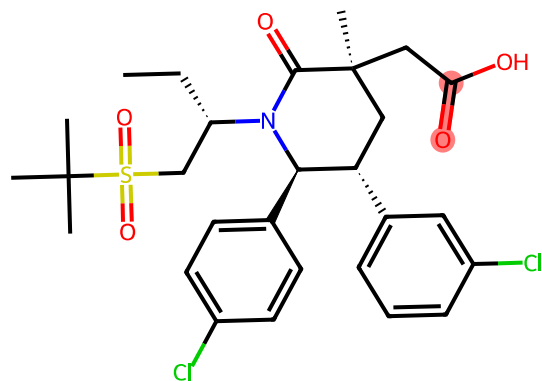


(A) biodolphin_Kd: prediction vs discovery**(B) biodolphin_Ki: prediction vs discovery****(C) biodolphin_Kd: protocol recall trajectories****(D) biodolphin_Ki: protocol recall trajectories**

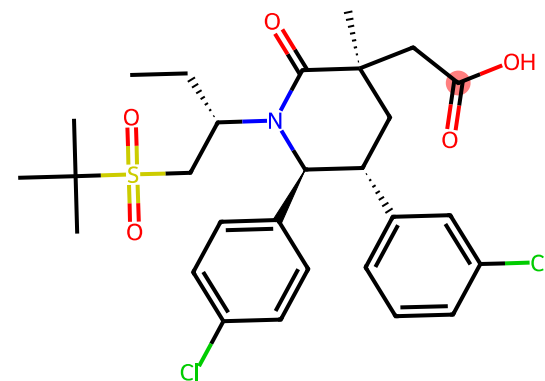
(A) biodolphin_Kd: MAE (lower)**(B) biodolphin_Kd: Recall AUC (higher)****(C) biodolphin_Ki: MAE (lower)****(D) biodolphin_Ki: Recall AUC (higher)**

(A) biodolphin_Kd: explore_heavy**(B) biodolphin_Kd: exploit_heavy****(C) biodolphin_Ki: explore_heavy****(D) biodolphin_Ki: exploit_heavy**

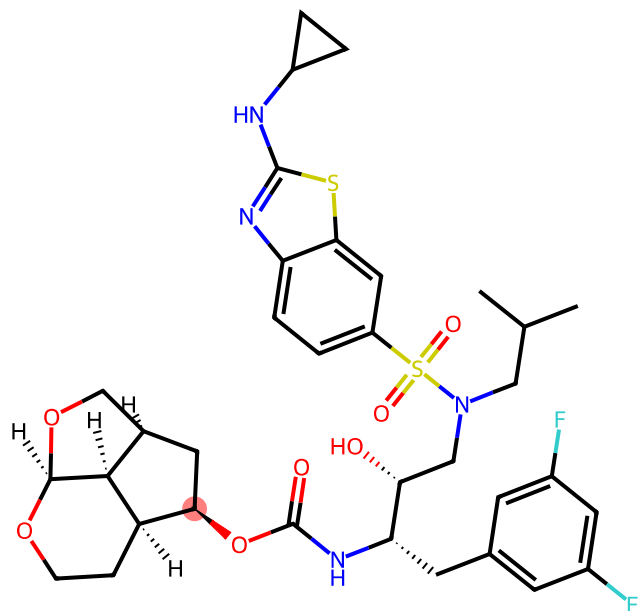
(A) biodolphin_Kd: selected-model attributions**(B) biodolphin_Ki: selected-model attributions****(C) ECFP-linear protocol stability****(D) Small panels: lowest-MAE configurations**



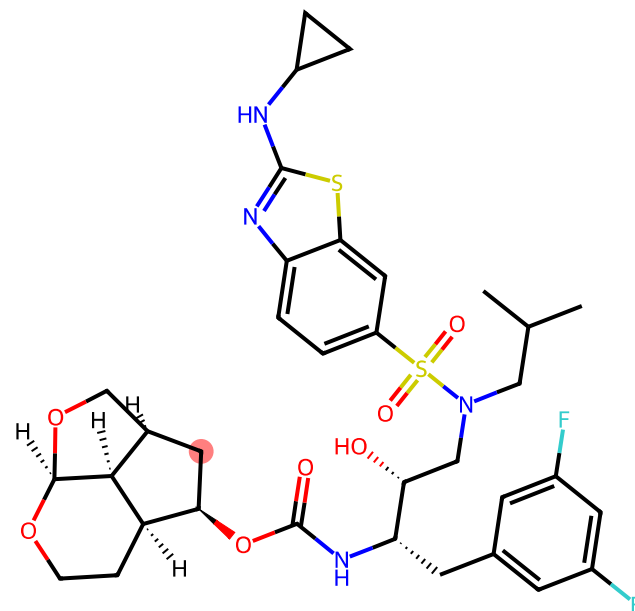
(A) biodolphin_Kd | bit 1917
Measured pK 10.39; retrospective example



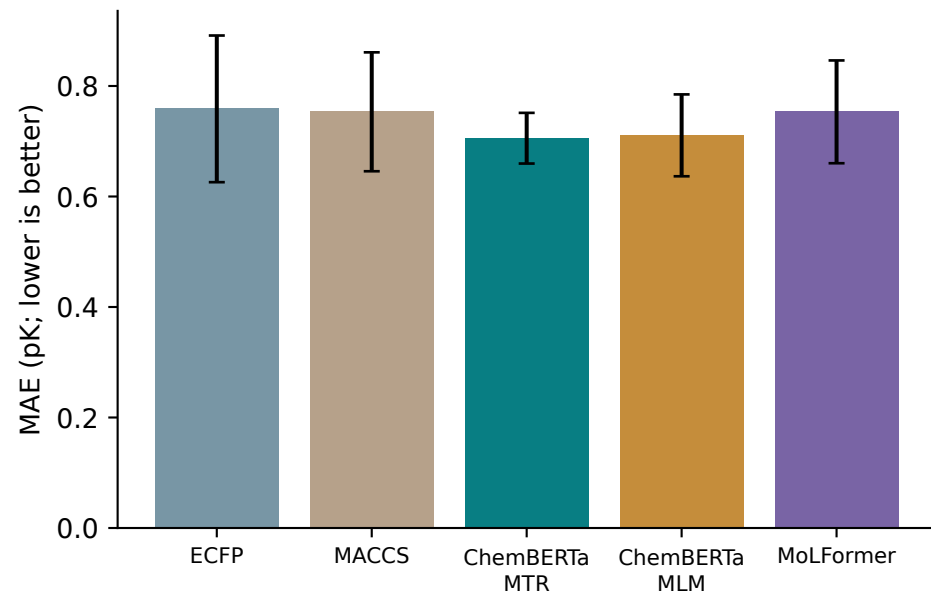
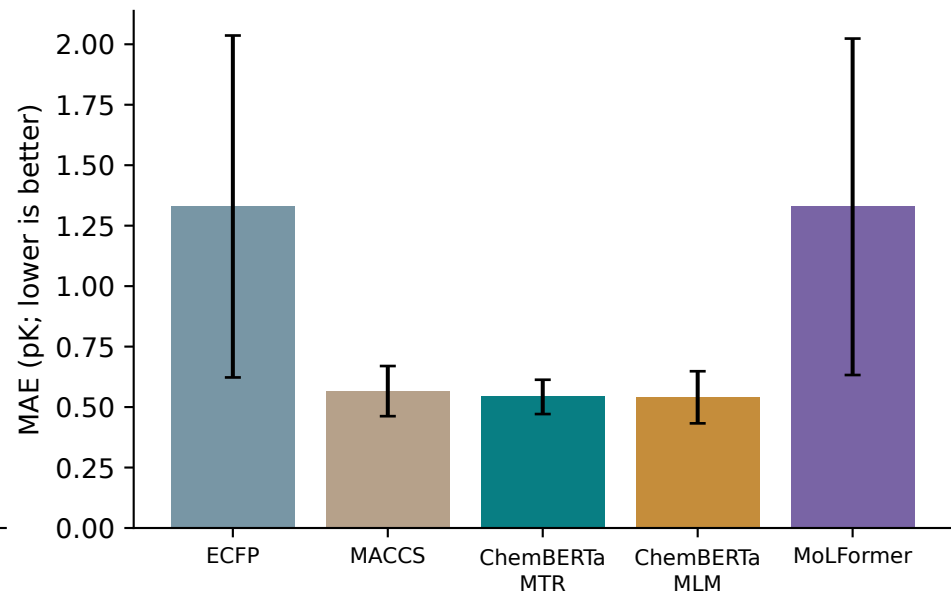
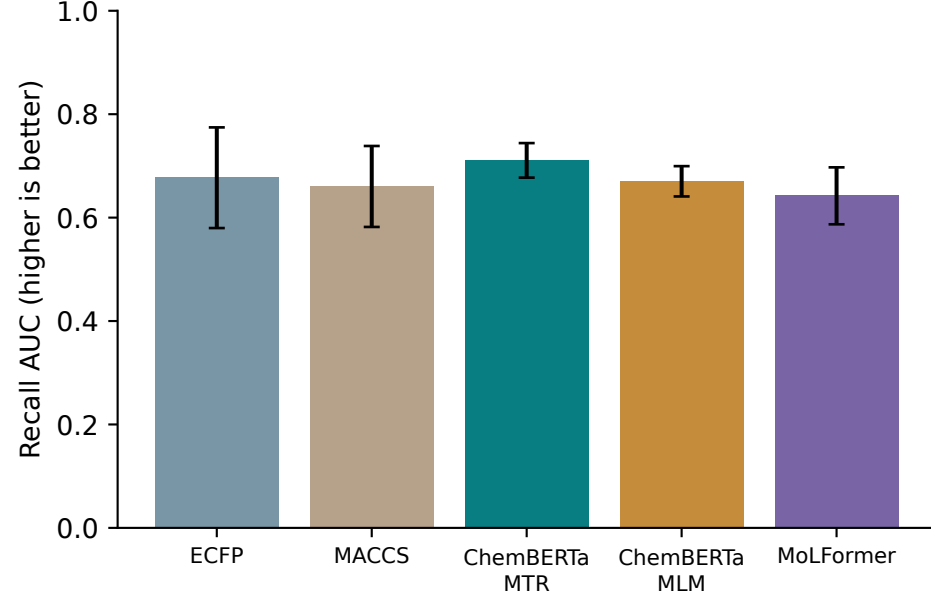
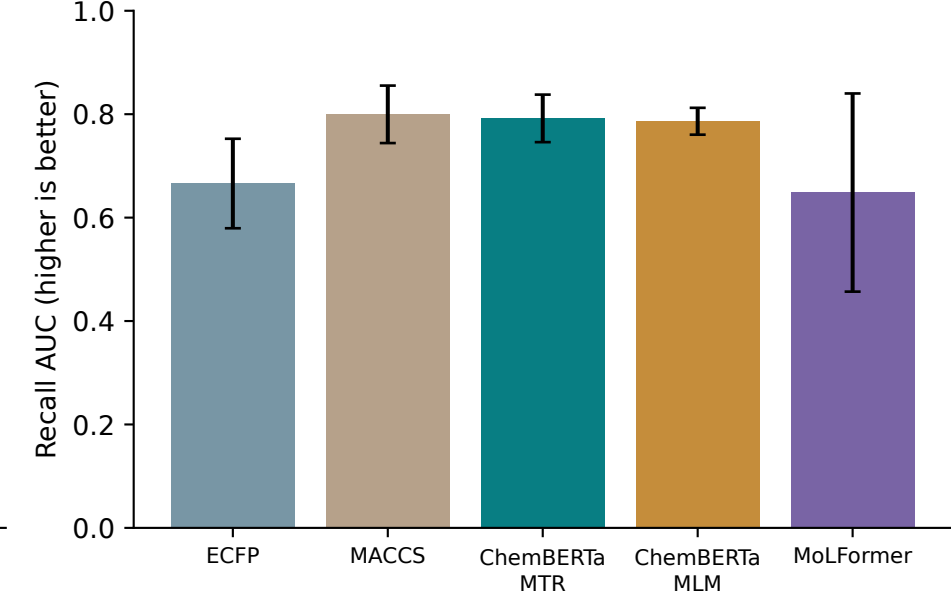
(B) biodolphin_Kd | bit 2855
Measured pK 10.39; retrospective example



(C) biodolphin_Ki | bit 3067
Measured pK 11.70; retrospective example



(D) biodolphin_Ki | bit 2974
Measured pK 11.70; retrospective example

(A) BioDolphin K_d (B) BioDolphin K_i (C) BioDolphin K_d (D) BioDolphin K_i 

(A) Experiment selection

Choose an experiment

Analysis

Full benchmark

Dataset / target

biodolphin Kd

Representation

ChemBERTa-MLM - 384D

Kernel

Matérn 3/2

Acquisition policy

Joint Thompson sampling

Seed

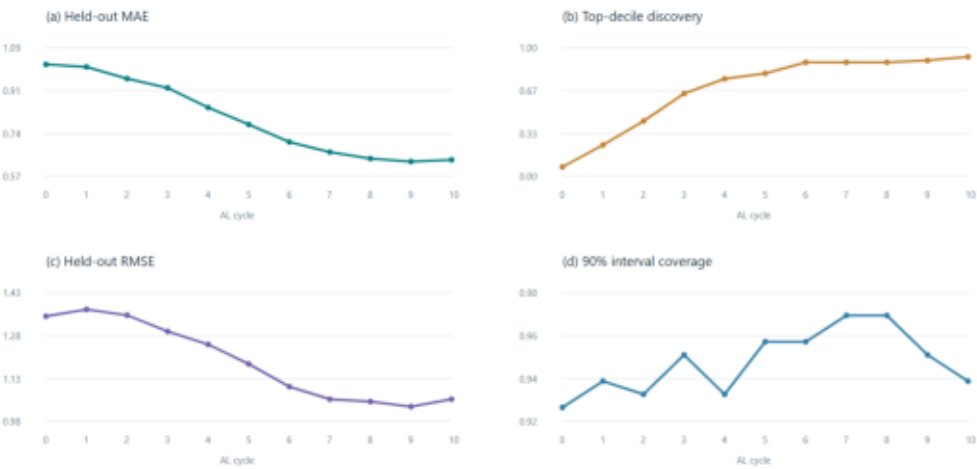
7

5850 completed benchmark runs - 90 explanation reports

(C) Matched representation comparison

Representation	Seeds	MAE $\pm$ SD	R ² $\pm$ SD	Top-10% recall $\pm$ SD	Recall AUC $\pm$ SD
ECFP - radius 4 / 4096 bits	3	0.759 $\pm$ 0.133	0.578 $\pm$ 0.103	0.924 $\pm$ 0.050	0.677 $\pm$ 0.097
MACCS - 167 bits	3	0.753 $\pm$ 0.108	0.565 $\pm$ 0.095	0.852 $\pm$ 0.139	0.660 $\pm$ 0.078
ChemBERTa-MTR - 384D	3	0.705 $\pm$ 0.046	0.624 $\pm$ 0.007	0.967 $\pm$ 0.016	0.711 $\pm$ 0.033
ChemBERTa-MLM - 384D	3	0.711 $\pm$ 0.074	0.620 $\pm$ 0.035	0.919 $\pm$ 0.030	0.670 $\pm$ 0.029
MolFormer-XL - 768D	3	0.753 $\pm$ 0.093	0.561 $\pm$ 0.060	0.843 $\pm$ 0.080	0.642 $\pm$ 0.055

(B) Four-panel learning curves



(D) Local upload campaign controls

1. Upload measured pairs

CSV file - 5 MB maximum - 10-1000 rows

Choose File No file chosen

Required columns: protein_sequence, lipid_smiles, affinity.

Use one affinity endpoint, standard amino-acid sequences and valid SMILES. Every affinity must be measured and numeric. Duplicates are rejected after canonicalizing SMILES.

2. Configure the experiment

Affinity units

pKd / pKi (larger is better)

Lipid representation

MolFormer-XL - 768D

Lipid kernel

tanimoto

Acquisition policy

Thompson + diversity heuristic

Initial measured pairs

10

Pairs acquired per cycle

5

Acquisition cycles

5

Random seed

7

Transformers run locally using downloaded frozen weights. EI/PI use latent uncertainty and $\xi_i = 0.01$ pK (marginal top-k batch, not qEI). Thompson + diversity uses a fixed 0.25 diversity bonus. All five lipid kernels are available for all five representations.

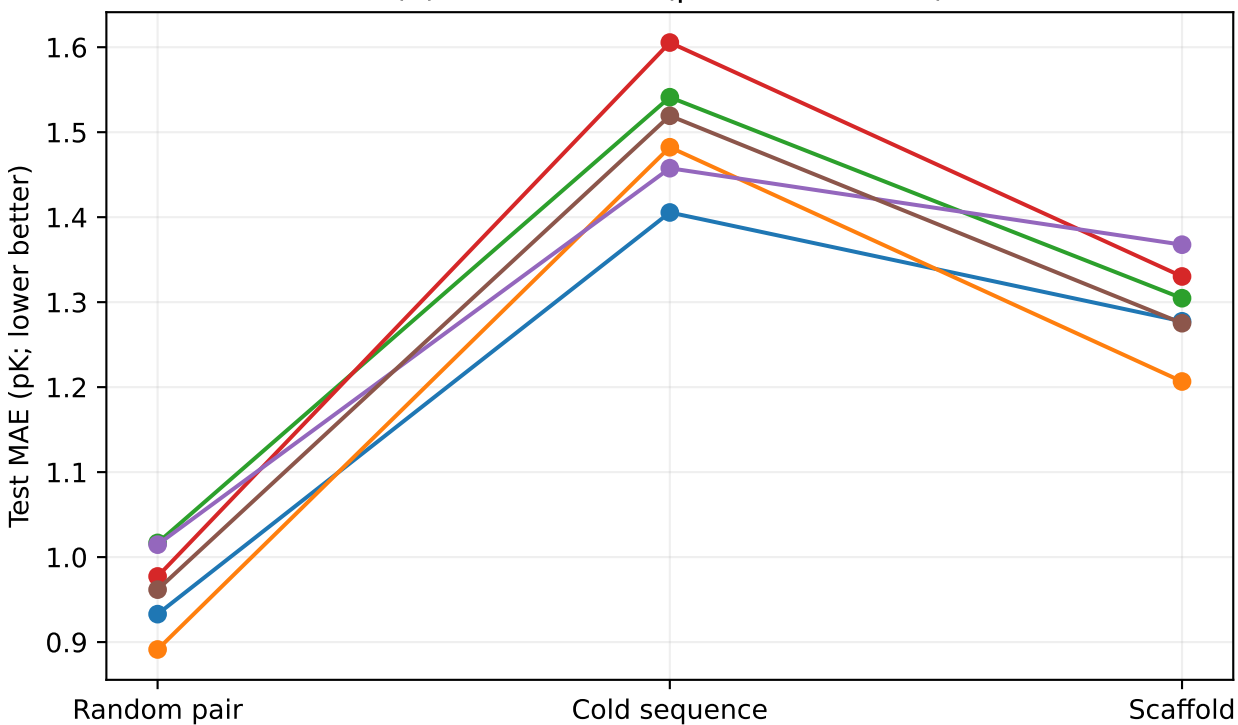
A frozen 20% random-pair test set is excluded from acquisition. Initial + batch $\times$ cycles must fit within the remaining pool (maximum 500 acquired labels). Raw Kd/Ki is converted to pK. Protein composition uses a Matérn 3/2 factor.

This is retrospective AL, not a live experimental oracle. Input data remain on this computer under results/user_runs. New runs produce metrics, query indices and test predictions; SHAP is not automatically computed for uploads.

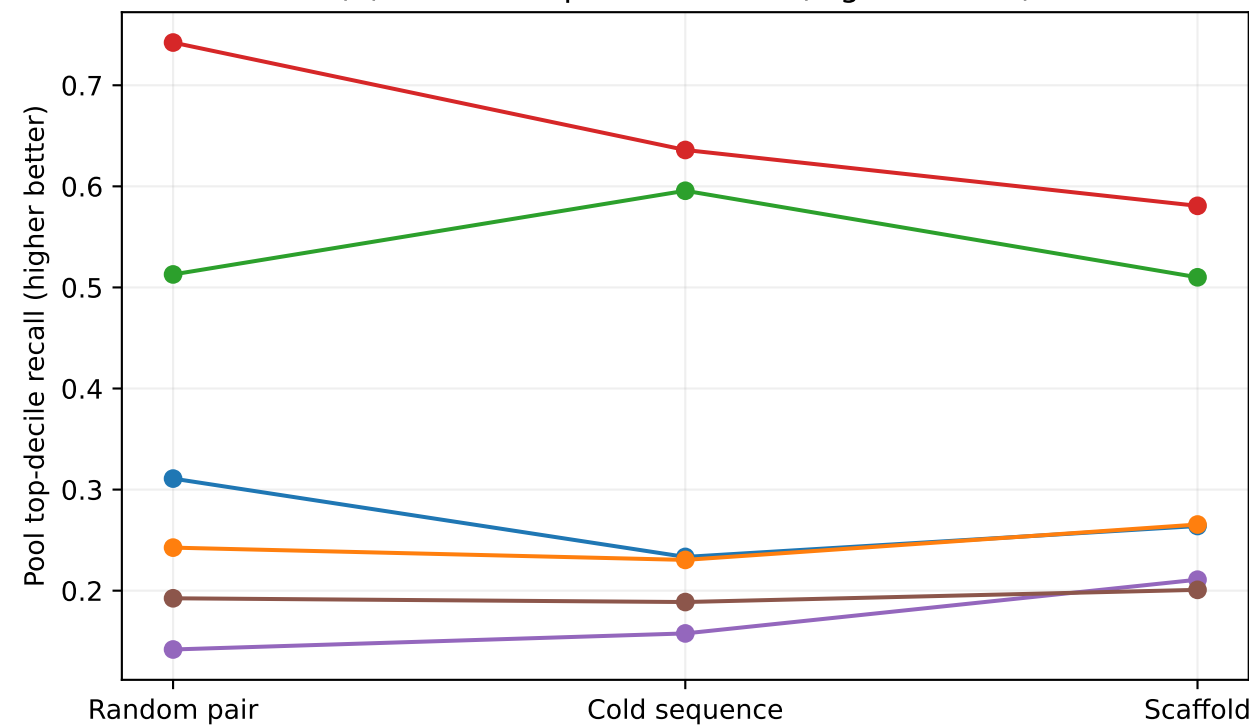
Validate & start campaign -- No benchmark results are overwritten.

Prediction versus discovery
Six-seed descriptive means; configurations chosen by inner validation

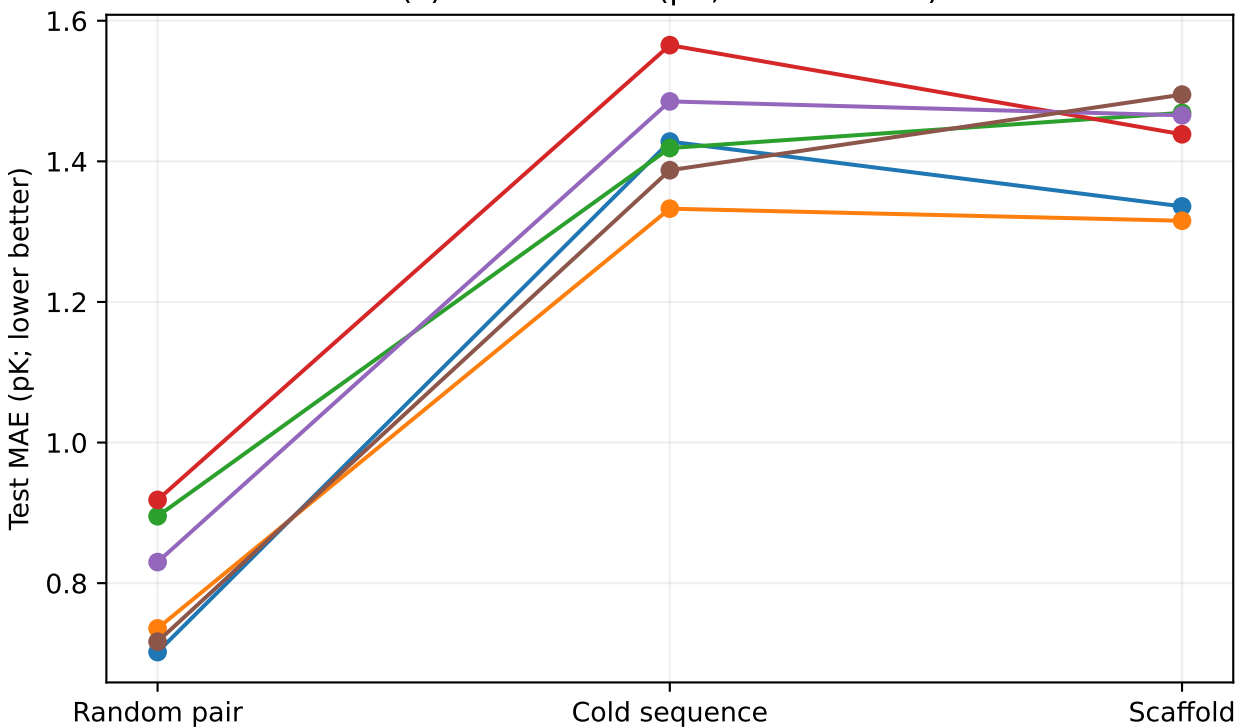
(a) Kd: Test MAE (pK; lower better)



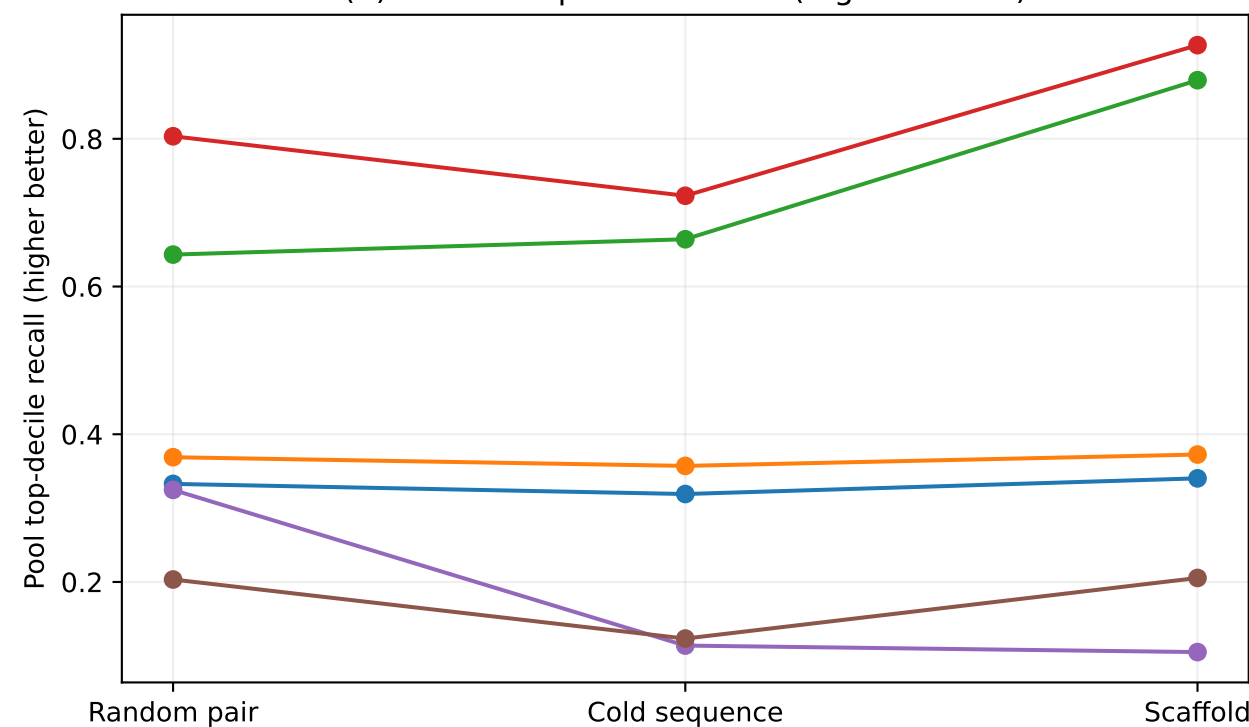
(b) Kd: Pool top-decile recall (higher better)



(c) Ki: Test MAE (pK; lower better)

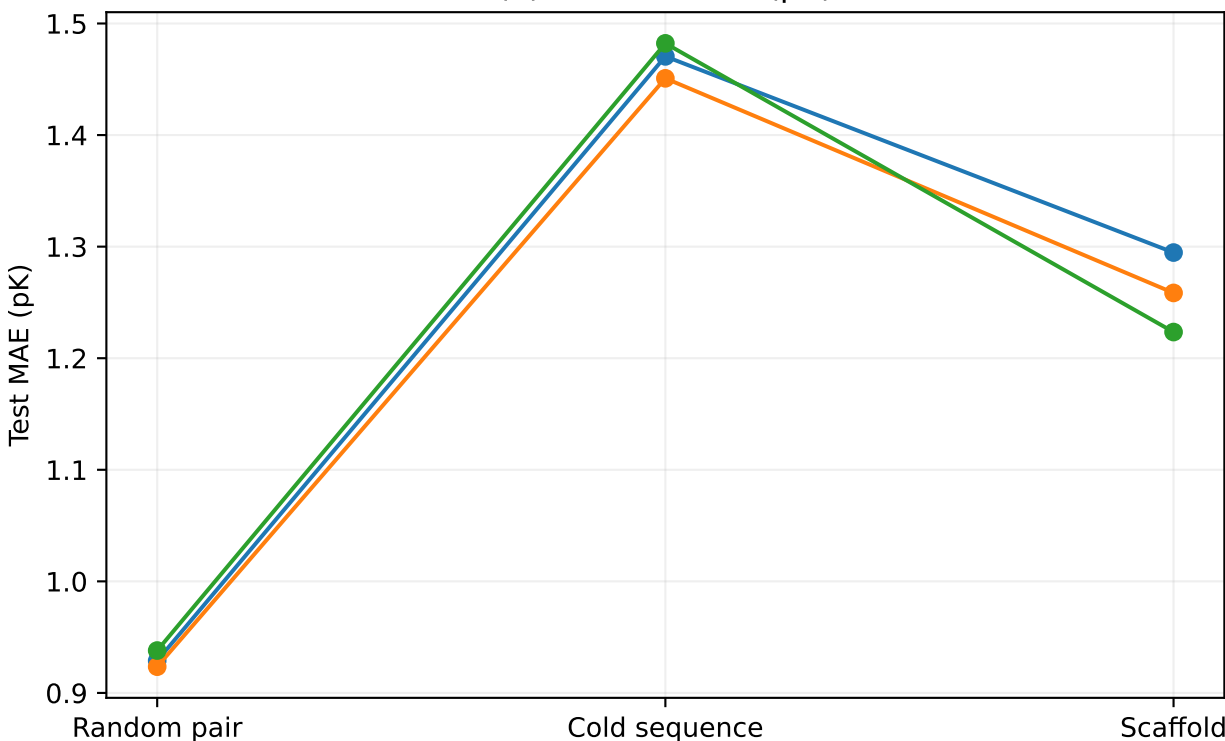


(d) Ki: Pool top-decile recall (higher better)

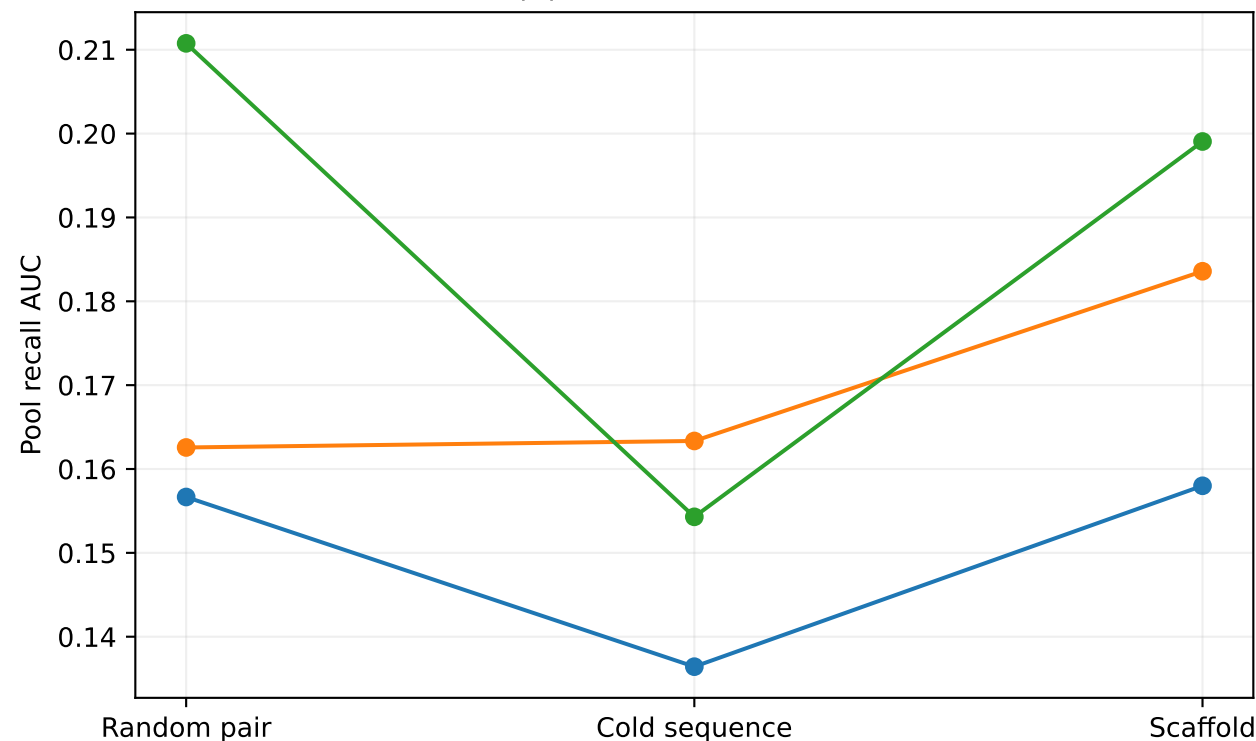


Protein representations: prediction-aware Thompson
Six-seed descriptive means; configurations chosen by inner validation

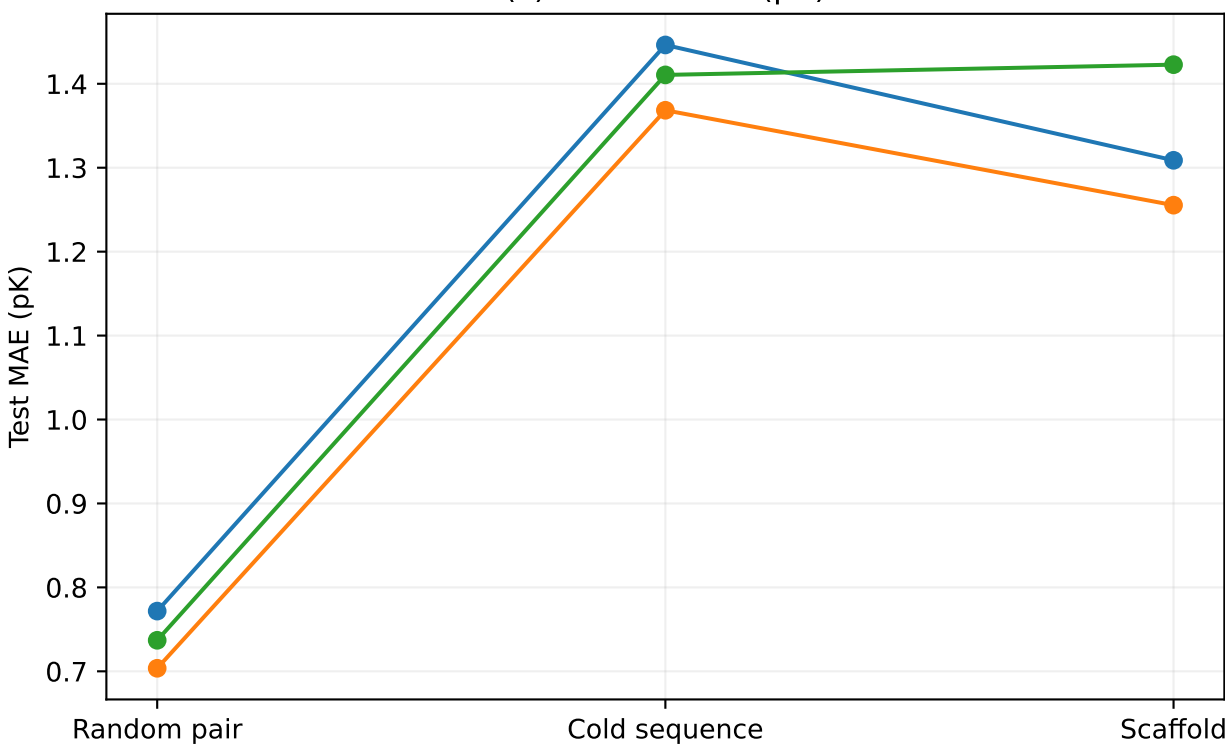
(a) Kd: Test MAE (pK)



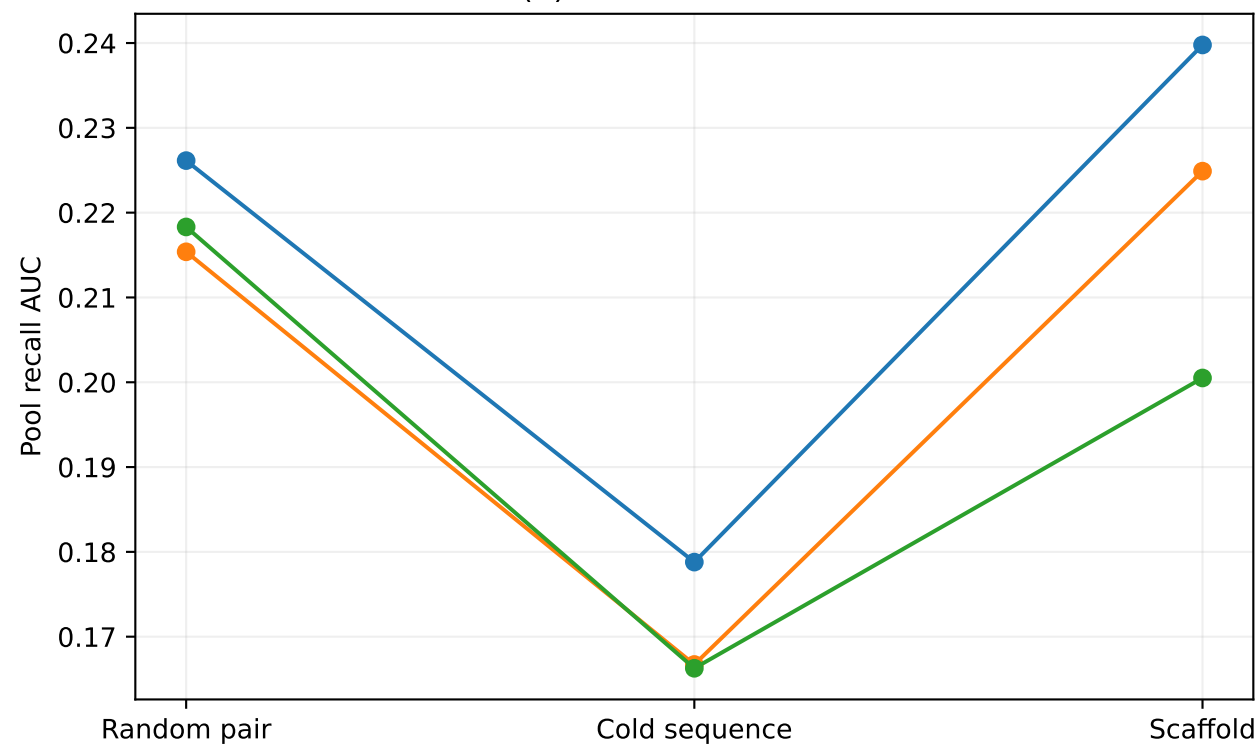
(b) Kd: Pool recall AUC



(c) Ki: Test MAE (pK)



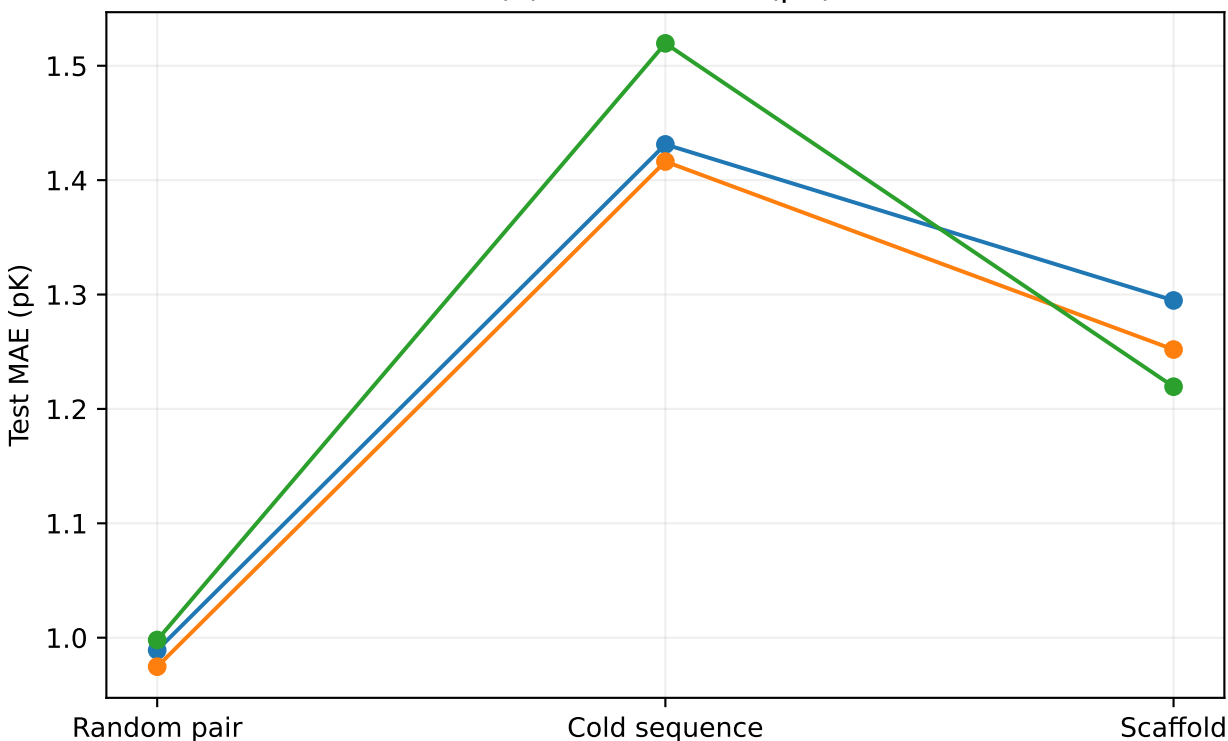
(d) Ki: Pool recall AUC



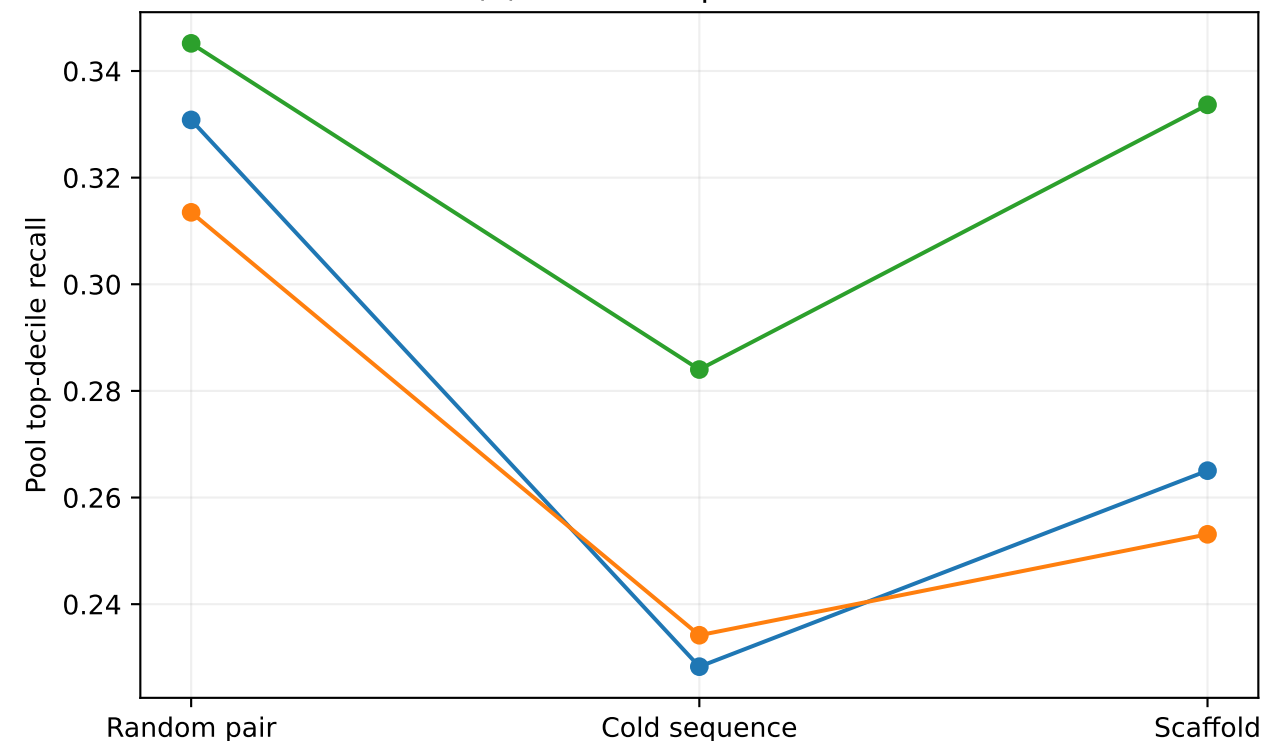
composition esm2 35m esm2 8m

Lipid adaptation: ESM-2 35M + ChemBERTa MLM
Six-seed descriptive means; configurations chosen by inner validation

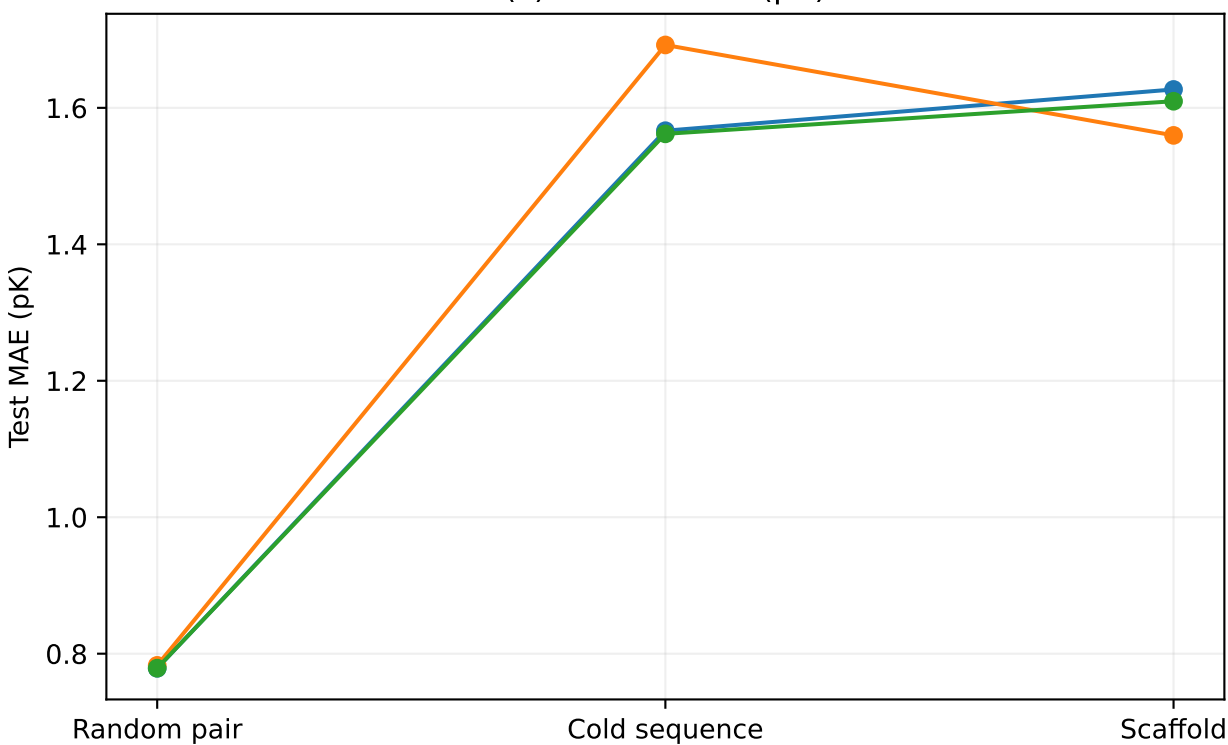
(a) Kd: Test MAE (pK)



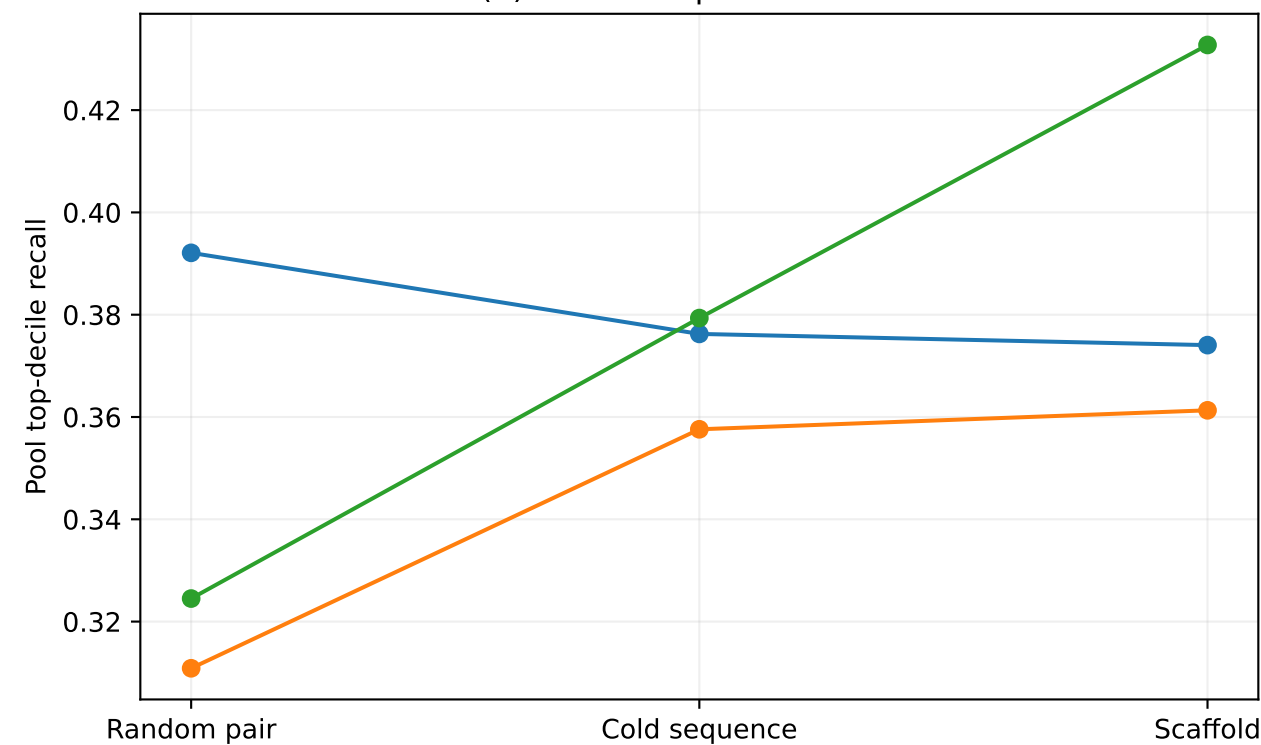
(b) Kd: Pool top-decile recall



(c) Ki: Test MAE (pK)



(d) Ki: Pool top-decile recall



—●— frozen mlm —●— initial label adapter —●— lipid ssl

(a) Dashboard navigation

LipidLab

PROTEIN · LIPID · DISCOVERY

Explore results

Run a campaign

Published measurements

Inspect saved explanations or evaluate your own measured pairs in a local active-learning campaign.

No synthetic affinities.

No remote upload service.

BENCHMARKS & EXPLAINABILITY

Compare how the model learns.

Explore the full representation, kernel and acquisition grid. Compare discovery with prediction, then inspect available explanations.

Saved, reproducible experiments

New: leakage-controlled study

11,016 additional unique campaigns: cold-sequence/scaffold splits, ESM-2 protein embeddings, prediction-aware Thompson and lipid adaptation.

Explore all new runs Updated full paper SHAP explanations Highlighted structures

New results use 120 acquired labels plus validation costs; older pooled runs below use 360. Public hosting is read-only.

Choose an experiment

Analysis

Full benchmark

Dataset / target

biodolphin Kd

Representation

ChemBERTa-MTR · 384D

Kernel

linear

Acquisition policy

Expected improvement (EI)

Seed

7

5850 completed benchmark runs · 90 explanation reports

Supported suite: 5 lipid representations × 5 kernels × 13 acquisition policies. Select "Full benchmark" to see every completed combination. Explanation views contain only separately computed SHAP reports.

0.730

Held-out MAE · pK

0.571

Held-out R²

71.4%

Top-decile pool recall

360

Labels · seed 7

Learning curves · selected seed

(a) Held-out MAE

(b) Top-decile discovery

(b) Highlighted structures

LipidLab

PROTEIN · LIPID · DISCOVERY

Explore results

Run a campaign

Published measurements

Inspect saved explanations or evaluate your own measured pairs in a local active-learning campaign.

No synthetic affinities.

No remote upload service.

Highlighted structures

Bit 2437 | pK 10.39

246/864 pairs

Bit 3108 | pK 10.39

412/864 pairs

Bit 1917 | pK 10.39

495/864 pairs

Bit 3612 | pK 9.21

284/864 pairs

(c) Leakage-controlled explorer

Explore every new configuration

Original benchmark / SHAP / highlighted structures · Full paper

9,720 main campaigns; 1,944 adaptation comparisons, including 648 reused baselines.

This browser exposes every run, including configurations not selected by validation. Do not select a model by its test score. The figures below use validation-only selections.

Study

Main: 9,720 campaigns

Endpoint

biodolphin Kd

Evaluation split

cold sequence40

Protein representation

esm2 35m

Lipid representation

chemberta mlm

Adaptation

frozen

Kernel

linear

Acquisition policy

prediction aware thompson

Seed

7

1.242

Test MAE (pK)

-0.283

Test R²

0.309

Pool recall

0.205

Recall AUC

1.429

Validation MAE

120 acquired labels; 257 total labels including validation. Nonconverged fits: 0. Frozen MLM adaptation entries reuse main-study baselines. No prospective evaluation.

► All configurations for the selected study, endpoint and split

Main per-seed CSV · Adaptation CSV · Paired tests · Split audit

(d) Adaptation explorer

Explore every new configuration

Original benchmark / SHAP / highlighted structures · Full paper

9,720 main campaigns; 1,944 adaptation comparisons, including 648 reused baselines.

This browser exposes every run, including configurations not selected by validation. Do not select a model by its test score. The figures below use validation-only selections.

Study

Adaptation: 1,944 comparison

Endpoint

biodolphin Kd

Evaluation split

cold sequence40

Protein representation

esm2 35m

Lipid representation

chemberta mlm

Adaptation

initial label adapter

Kernel

linear

Acquisition policy

prediction aware thompson

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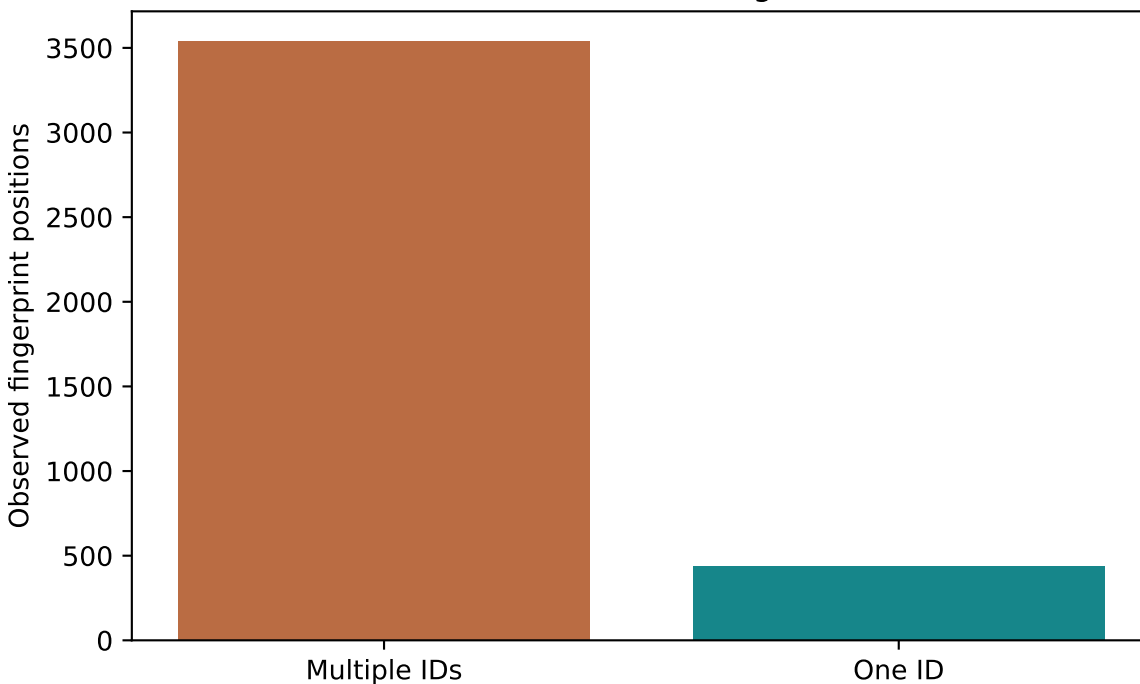
Validation MAE

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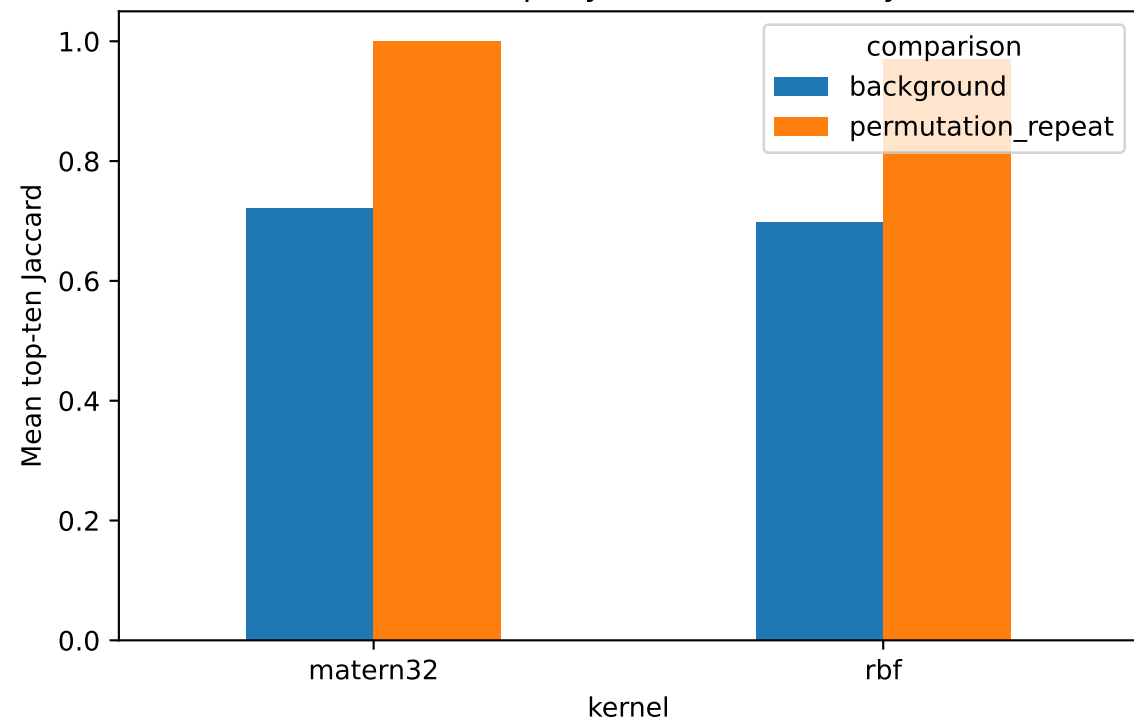
► All configurations for the selected study, endpoint and split

Main per-seed CSV · Adaptation CSV · Paired tests · Split audit

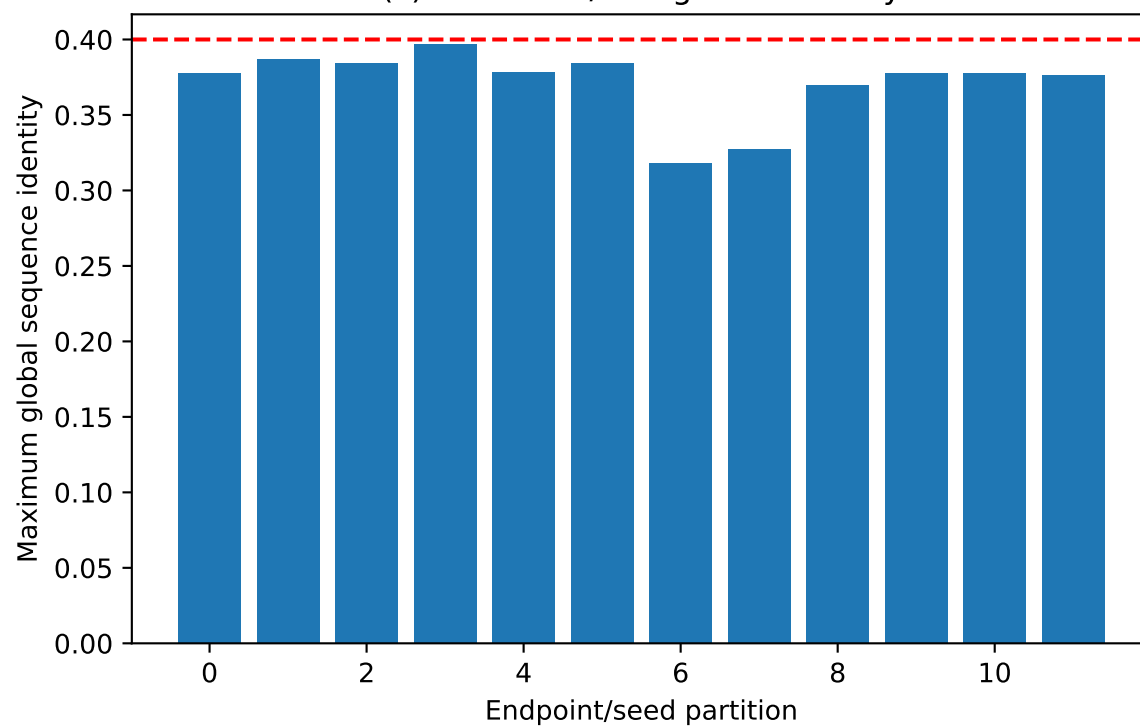
(a) Observed ECFP folding collisions



(b) Small-query nonlinear stability



(c) Cold train/test global identity



(d) Four examples; only two complexes

